AMRITA VISHWA VIDYAPEETHAM
DEEMED TO BE UNIVERSITY UNDER SECTION 3 OF UGC ACT, 1956

SCHOOL OF COMPUTING CHENNAI

DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING

23CSE304 Embedded Systems Laboratory

Activity Sheet #1

Activity Name: GPIO LED Interfacing

Name of Student:	**Chetan Kumar G**
Registration No.:	**CH.SC.U4CSE24109**
Activity Date:	**28|07|2026**
Submission Date:	**28|07|2026**

Assessment Rubrics:

Component	Maximum Marks	Marks Allotted
Program	3	
Output	4	
Viva	2	
Timely Submission	1	
Total	**10**	

Faculty In-charge

Write an embedded C program for ARM architecture to interface an LED. Debug and verify the program using IDE on a development computer system.

PROGRAM:

```
//CH.SC.U4CSE24109
#include <LPC21xx.h>

void delay(void);

int main()
{
    IO0DIR |= (1 << 0);       // Set P0.0 as Output

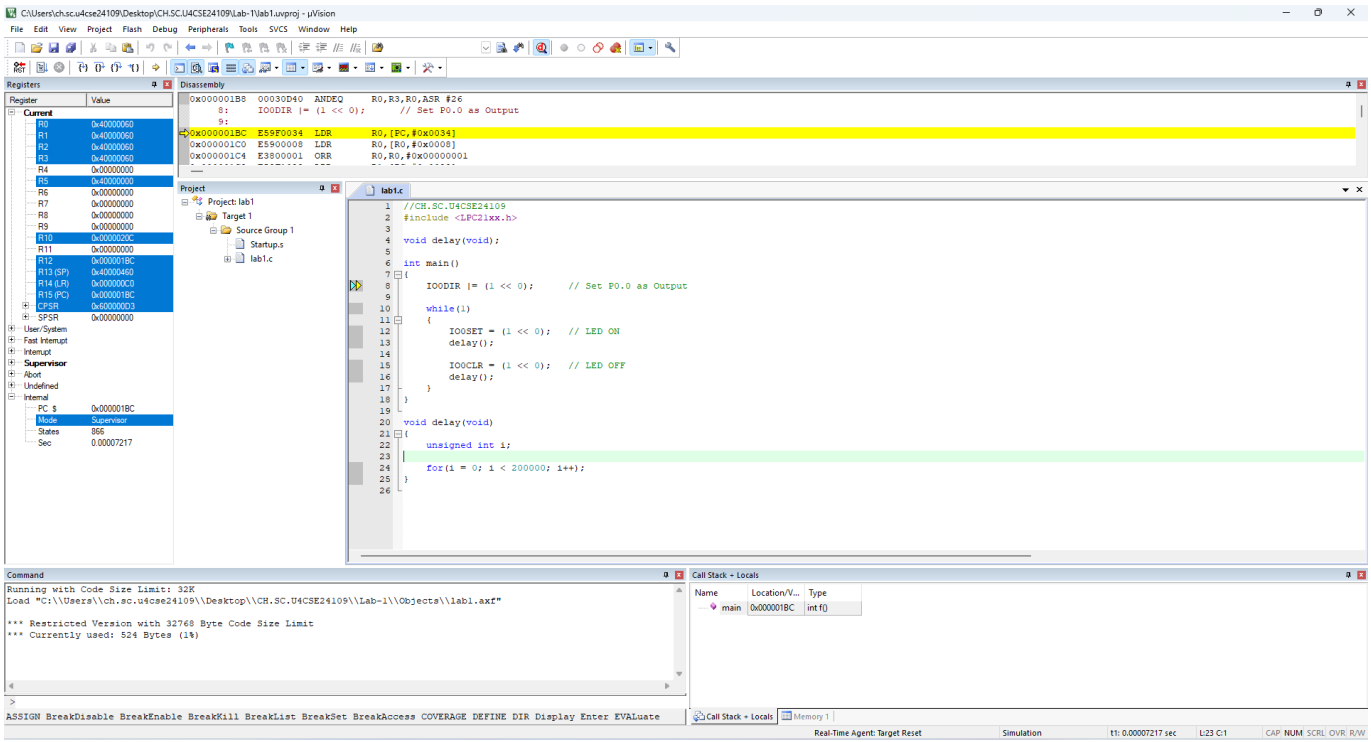
    while(1)
    {
        IO0SET = (1 << 0);    // LED ON
        delay();

        IO0CLR = (1 << 0);    // LED OFF
        delay();
    }
}

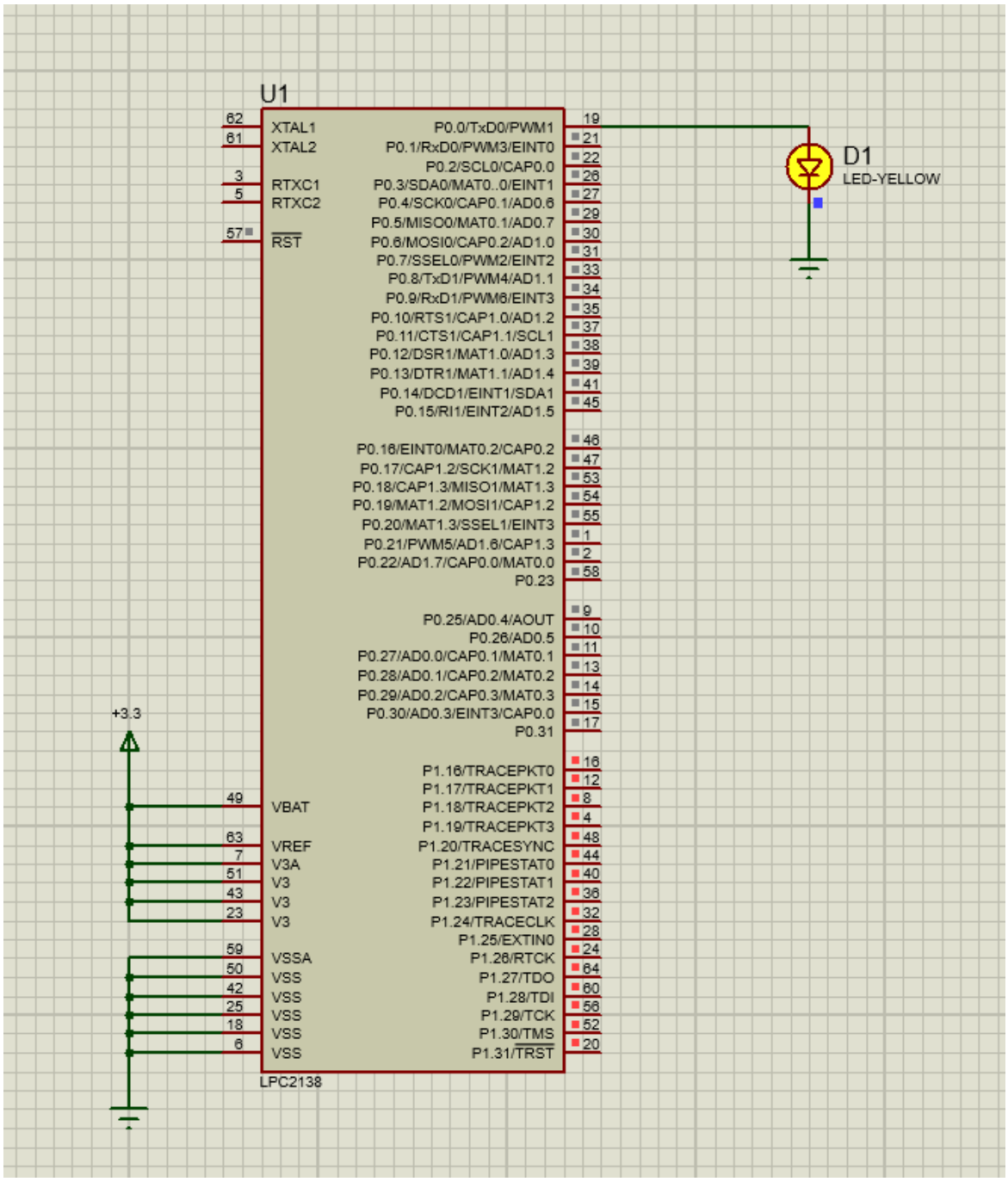
void delay(void)
{
    unsigned int i;

    for(i = 0; i < 200000; i++);
}
```

OUTPUT WINDOW:


C:\Users\ch.sc.u4cse24109\Desktop\CH.SC.U4CSE24109\Lab-1\lab1.uvproj - µVision
File Edit View Project Flash Debug Peripherals Tools SVCS Window Help
Registers
Register Value
Current
R0 0x40000060
R1 0x40000060
R2 0x40000060
R3 0x40000060
R4 0x00000000
R5 0x40000000
R6 0x00000000
R7 0x00000000
R8 0x00000000
R9 0x00000000
R10 0x000028BC
R11 0x00000000
R12 0x000001BC
R13 (SP) 0x40000460
R14 (LR) 0x000000C0
R15 (PC) 0x000001BC
CPSR 0x60000D3
SPSR 0x00000000
User/System
Fast Interrupt
Interrupt
Supervisor
Abort
Undefined
Internal
PC $ 0x000001BC
Mode Supervisor
States 866
Sec 0.00007217
Disassembly
0x000001B8 00030D40 ANDEQ R0,R3,R0,ASR #26
8: IO0DIR |= (1 << 0); // Set P0.0 as Output
9:
0x000001BC E59F0034 LDR R0,[PC,#0x0034]
0x000001C0 E5900008 LDR R0,[R0,#0x0008]
0x000001C4 E3800001 ORR R0,R0,#0x00000001
Project
Project: lab1
Target 1
Source Group 1
Startup.s
lab1.c
lab1.c
1 //CH.SC.U4CSE24109
2 #include <LPC21xx.h>
3
4 void delay(void);
5
6 int main()
7 {
8 IO0DIR |= (1 << 0); // Set P0.0 as Output
9
10 while(1)
11 {
12 IO0SET = (1 << 0); // LED ON
13 delay();
14
15 IO0CLR = (1 << 0); // LED OFF
16 delay();
17 }
18 }
19
20 void delay(void)
21 {
22 unsigned int i;
23
24 for(i = 0; i < 200000; i++);
25 }
26
Command
Running with Code Size Limit: 32K
Load "C:\\Users\\ch.sc.u4cse24109\\Desktop\\CH.SC.U4CSE24109\\Lab-1\\Objects\\lab1.axf"
*** Restricted Version with 32768 Byte Code Size Limit
*** Currently used: 524 Bytes (1%)
ASSIGN BreakDisable BreakEnable BreakKill BreakList BreakSet BreakAccess COVERAGE DEFINE DIR Display Enter EVALuate
Call Stack + Locals
Name Location/V... Type
main 0x000001BC int f()
Call Stack + Locals
Memory 1
Real-Time Agent: Target Reset
Simulation
t1: 0.00007217 sec
L:23 C:1
CAP NUM SCRL OVR R/W

OUTPUT:


U1
62 XTAL1
61 XTAL2
3 RTXC1
5 RTXC2
57 RST
49 VBAT
63 VREF
7 V3A
51 V3
43 V3
23 V3
59 VSSA
50 VSS
42 VSS
25 VSS
18 VSS
6 VSS
P0.0/TxD0/PWM1 19
P0.1/RxD0/PWM3/EINT0 21
P0.2/SCL0/CAP0.0 22
P0.3/SDA0/MAT0..0/EINT1 26
P0.4/SCK0/CAP0.1/AD0.6 27
P0.5/MISO0/MAT0.1/AD0.7 29
P0.6/MOSI0/CAP0.2/AD1.0 30
P0.7/SSEL0/PWM2/EINT2 31
P0.8/TxD1/PWM4/AD1.1 33
P0.9/RxD1/PWM6/EINT3 34
P0.10/RTS1/CAP1.0/AD1.2 35
P0.11/CTS1/CAP1.1/SCL1 37
P0.12/DSR1/MAT1.0/AD1.3 38
P0.13/DTR1/MAT1.1/AD1.4 39
P0.14/DCD1/EINT1/SDA1 41
P0.15/RI1/EINT2/AD1.5 45
P0.16/EINT0/MAT0.2/CAP0.2 46
P0.17/CAP1.2/SCK1/MAT1.2 47
P0.18/CAP1.3/MISO1/MAT1.3 53
P0.19/MAT1.2/MOSI1/CAP1.2 54
P0.20/MAT1.3/SSEL1/EINT3 55
P0.21/PWM5/AD1.6/CAP1.3 1
P0.22/AD1.7/CAP0.0/MAT0.0 2
P0.23 58
P0.25/AD0.4/AOUT 9
P0.26/AD0.5 10
P0.27/AD0.0/CAP0.1/MAT0.1 11
P0.28/AD0.1/CAP0.2/MAT0.2 13
P0.29/AD0.2/CAP0.3/MAT0.3 14
P0.30/AD0.3/EINT3/CAP0.0 15
P0.31 17
P1.16/TRACEPKT0 16
P1.17/TRACEPKT1 12
P1.18/TRACEPKT2 8
P1.19/TRACEPKT3 4
P1.20/TRACESYNC 48
P1.21/PIPESTAT0 44
P1.22/PIPESTAT1 40
P1.23/PIPESTAT2 36
P1.24/TRACECLK 32
P1.25/EXTIN0 28
P1.26/RTCK 24
P1.27/TDO 64
P1.28/TDI 60
P1.29/TCK 56
P1.30/TMS 52
P1.31/TRST 20
LPC2138
+3.3
D1
LED-YELLOW